STUR Resistance Equation — Derivation Record

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Summary

Resistance is one loop phase, Φ_R, whose variance is entropy, whose energy cost is set by temperature, and whose ceiling is set by boundary area. The candidate equation:

$$\Phi_R = \oint_C A_\mu\, dx^\mu, \qquad \langle W(C) \rangle = e^{-\frac{1}{2}\langle \Phi_R^2 \rangle}$$

$$S_R = \tfrac{1}{2} k_B \Phi_R^2, \qquad E_R = \tfrac{1}{2} k_B T\, \Phi_R^2, \qquad \tfrac{1}{2}\Phi_R^2 = \sigma(\mu)\, A \le \frac{A}{4\ell_P^2}$$

One line of it is a STUR postulate: $S_R = ½\, k_B\, \Phi_R^2$. Everything else is either established physics or follows from that postulate, and every computed number below is reproduced by the companion script stur_verify.py.

Result	Status	Check
Visibility = Gaussian Wilson loop	Known physics	Monte Carlo agrees to 3 decimals
$S_R = ½\, k_B\, \Phi_R^2$	STUR postulate	Must be tested by experiment
$E_R = ½\, k_B\, T\, \Phi_R^2$	Follows (Landauer)	—
Ceiling reproduces black hole entropy and Smarr's $Mc^2 = 2TS$	Consistency check, passed	$E_R/Mc^2 = 0.500000$ for all masses tested
$½\Phi_R^2 = \sigma(\mu)A$ spans QCD to Planck	STUR hypothesis	Running of σ(μ) not yet derived
A fold resists collapse at a definite size	Known physics (Skyrme)	B=1 energy 1.2317 × 12π², literature 1.232

Status legend and physical picture

Each step is labelled so a referee can see exactly where STUR adds something new:

- **Known physics:** established, published, independently tested.
- **STUR postulate:** a new claim the theory makes; it stands or falls by experiment.
- **Follows:** derived from the lines above it with no new assumption.
- **Consistency check:** a place where the equation must reproduce a known exact result, and does or does not.

The picture being formalised. All matter ends in black holes, which evaporate by Hawking radiation until the universe holds almost no energy and almost no information. That empty state is the condition for a universe-wide quantum tunneling event, with no parent vacuum. Spacetime resists that tunneling, and the resistance appears as quarks: folds of spacetime that hold a definite size and carry spin. Folds bind into larger structures, and the cycle repeats.

Two working axioms carry the picture into equations: the universe is holographic (a region's information lives on its boundary), and energy and information are interconvertible at a stated exchange rate.

Step 1 — Resistance holonomy

Known physics. Φ_R is the phase picked up around a closed loop C by a gauge connection A_μ. It must be a connection, not a current, for the loop integral to be gauge invariant:

$$\Phi_R = \oint_C A_\mu\, dx^\mu = \int_{\Sigma} F_{\mu\nu}\, d\Sigma^{\mu\nu} \quad (\partial \Sigma = C)$$

The second equality is Stokes' theorem: the loop phase equals the field flux through any surface the loop bounds. That surface is where area, and so holography, enters later.

Visibility measures phase spread, not phase. A fixed phase shifts interference fringes but leaves visibility at 1. Visibility drops only when the phase fluctuates. For Gaussian fluctuations with variance $\langle \Phi_R^2 \rangle$:

$$V = V_0\, \big|\langle e^{i\Phi_R}\rangle\big| = V_0\, e^{-\frac{1}{2}\langle \Phi_R^2 \rangle}$$

The script checks this by Monte Carlo with 2 million samples: at σ = 1 rad, V = 0.6063 against e^(−½) = 0.6065. So the STUR visibility formula $V = V_0 \exp(-\Phi_R^2/2)$ is exact provided Φ_R^2 is read as the variance of the resistance phase.

Open choice. Which connection A_μ is: the spin connection (ties to spin ½ and torsion, so TEGR), U(1), or a new field. The rest of this record holds for any of them.

Step 2 — Resistance entropy (the postulate)

STUR postulate. The loss of visibility is entropy recorded by spacetime:

```
S_R \equiv -k_B \ln\frac{V}{V_0} = \tfrac{1}{2} k_B\, \Phi_R^2
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This is the single new claim in the equation. Its support: visibility loss is tied to information leaked about which path a particle took, through the proven duality $V^2 + D^2 \leqslant 1$ (D = path distinguishability). The postulate goes further and fixes the entropy as exactly −ln V in units of k_B.

What a referee will ask. For ordinary decoherence by a known environment, the entropy produced is not in general −ln V. The postulate is therefore a claim that spacetime itself acts as a specific environment. It must be tested, not assumed (see Open problems).

Step 3 — Energy, the holographic ceiling, and the Smarr check

Follows (Landauer). Recording entropy S at temperature T costs at least E = TS:

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E_R = \tfrac{1}{2} k_B T\, \Phi_R^2
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This has the form of equipartition: Φ_R behaves as one thermal quadratic mode carrying $\frac{1}{2}k_BT$. It also matches the form $E = \frac{1}{2}Nk_BT$ that Verlinde used to derive Newtonian gravity from a holographic screen.

Follows (Bekenstein / holography). No region holds more entropy than a black hole with the same boundary area, $S \leqslant k_B A / 4\ell_P^2$. So:

```
\Phi_R^2 \le \frac{A}{2\ell_P^2}
```

Consistency check — passed. Saturate the ceiling for a Schwarzschild black hole of mass M and set T to its Hawking temperature. Then S_R equals the Bekenstein–Hawking entropy, and

$$E_R = T_H S_{BH} = \frac{\hbar c^3}{8\pi G M k_B}\cdot \frac{4\pi G M^2 k_B}{\hbar c} = \tfrac{1}{2} M c^2$$

That is Smarr's exact relation $Mc^2 = 2TS$, reproduced without being put in. The script confirms $E_R/Mc^2 = 0.500000000000$ for masses from 10^{-3} to 10^{11} solar masses. At one solar mass the ceiling is $\Phi_R^2 \approx 2.1 \times 10^{77}$.

Step 4 — From quantum mechanics to quantum field theory

Known physics. QM is the low-energy, fixed-particle-number limit of QFT, so the equation must hold in both forms. In QM, Φ_R is the phase difference between two paths of one particle. In QFT the loop becomes the Wilson loop operator, and for a Gaussian (free) field its expectation value is exact:

$$\langle W(C) \rangle = \exp\!\Big(-\tfrac{1}{2}\langle \Phi_R^2 \rangle\Big), \qquad \langle \Phi_R^2 \rangle = g^2 \oint_C\!\oint_C dx^\mu\, dy^\nu\, D_{\mu\nu}(x-y)$$

$D_{\mu\nu}$ is the field's propagator. The interferometer formula of Step 1 is this QFT result evaluated for one loop, so the same Φ_R serves both theories.

How the phase variance grows tells the phase of the theory:

Behaviour of $\frac{1}{2}\langle\Phi_R^2\rangle$	Name	Physical meaning	Example
$\propto$ perimeter of C	Perimeter law	Charges are free	Electromagnetism
$= \sigma \times$ area of C	Area law	Charges are confined	QCD (quarks)

The area law is the standard signature of quark confinement, with σ the string tension.

Step 5 — One area law from quarks to black holes

STUR hypothesis. The QCD area law and the holographic ceiling of Step 3 have the same form, a tension times an area. STUR claims they are one law with a scale-dependent tension:

$$\tfrac{1}{2}\Phi_R^2 = \sigma(\mu)\, A$$

Scale μ	Tension σ	Value (m^{-2})	Describes
QCD	$(0.44\ \text{GeV})^2 \approx 0.98\ \text{GeV/fm}$	5.0×10^{30}	Quark confinement
Planck	$1/(4\ell_P^2)$	9.6×10^{68}	Black hole entropy, the holographic maximum

The two tensions differ by a factor of 1.9×10^{38}. This matches the physical claim that quark binding and gravitational collapse are two levels of one resistance.

What is not yet done. No toy flow is given here, because any curve drawn between the two endpoints without a derived equation would be decoration. What a real result needs is a flow equation

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\frac{d\sigma}{d\ln\mu} = \beta(\sigma)
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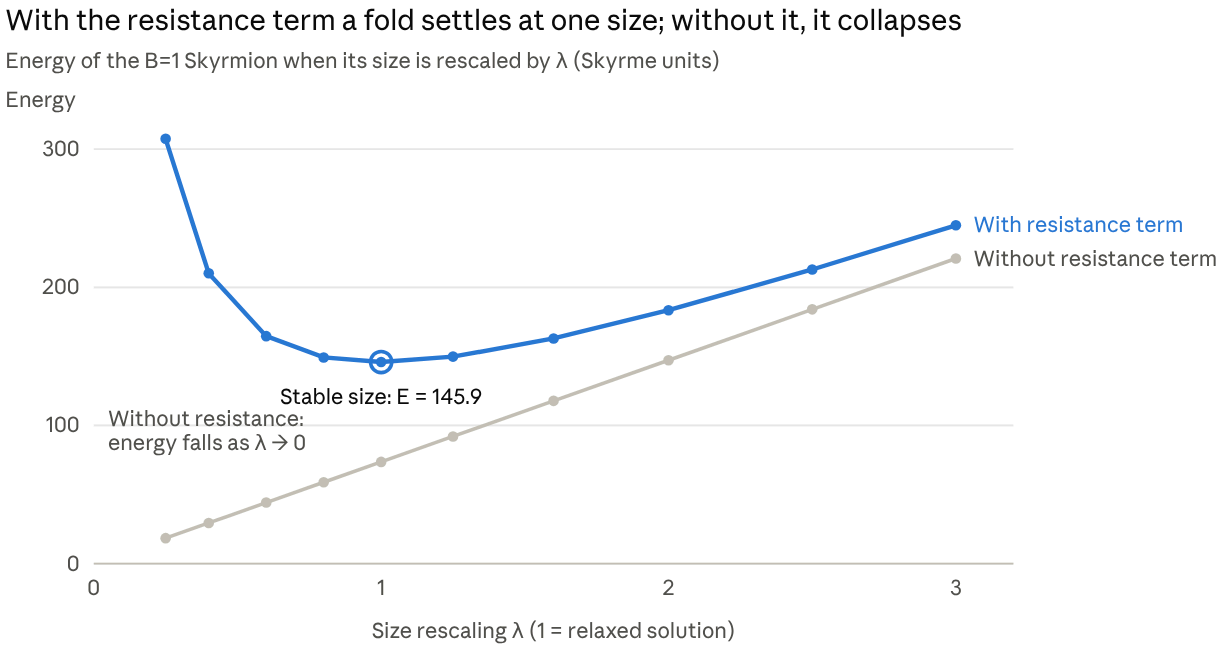
derived from the theory, which carries σ from one endpoint to the other and keeps electromagnetism on the perimeter law in between. In holographic language, μ is the depth of the bulk (holographic renormalisation), so this is the calculation of how Φ_R changes with depth. This step is where QFT meets gravity.

Step 6 — Resistance to collapse

Known physics. Derrick's theorem: any static lump of an ordinary field in 3D lowers its energy by shrinking, so it collapses. Rescale a lump's size by λ and its two energy terms scale oppositely:

```
E(\lambda) = \lambda E_2 + \frac{E_4}{\lambda}, \qquad \lambda_* =
\sqrt{E_4/E_2}, \qquad E_{\min} = 2\sqrt{E_2 E_4}
```

E_2 (gradient energy) drives collapse. E_4 (Skyrme's term) grows as the lump shrinks: it is a precise mathematical form of "resistance to collapse".



stur_verify.py, check 5 · relaxed B=1 hedgehog, 4,000-point radial grid

Numerical results (script, check 5). The relaxed solutions reproduce the published values, so the solver is trustworthy:

Configuration	Energy / (12π² × B)	Published
One fold (B=1, exact hedgehog)	1.2317	1.232
Two folds merged (B=2, rational-map ansatz)	1.2083	1.208

Two merged folds have 5.55 units less energy than two separate ones (1.9%): combining makes them more tightly bound, as the STUR picture requires. At the minimum $E_2 \approx E_4$ (73.6 vs 72.3, equal up to grid error), as Derrick's virial theorem demands.

Spin. Quantised Skyrmions carry spin ½ when the number of colours is odd (Witten, 1983); pure-geometry lumps (geons) can also carry spin ½ (Friedman–Sorkin, 1980). In holonomy terms, spin ½ means a 2π loop returns the state multiplied by −1. STUR's task is a resistance term built from spacetime geometry, not from a separate field, that does what E_4 does here.

The cosmological cycle

Evaporation is slow but finite (known physics). Hawking evaporation time is $t = 5120\pi G^2M^3/(\hbar c^4)$:

Black hole	Evaporation time
1 solar mass	2.1×10^{67} yr
Sgr A* (4×10^{6} solar masses)	1.3×10^{87} yr
10^{11} solar masses (largest known class)	2.1×10^{100} yr
All the observable universe's mass in one hole	9.0×10^{135} yr

Accelerating expansion means matter ends in many separate black holes, not one. Each still evaporates, so the cycle only needs "all black holes", not "a single black hole".

Two resistances, one candidate quantity. The cycle contains two collapses spacetime could resist:

1. **Matter collapse.** The loop-quantum-cosmology form is the template (known physics in that framework): the bracket stops collapse at $\rho_c \approx 0.41\,\rho_{Planck}$.

```
H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right)
```

2. **The tunneling event.** The rate grows as the universe empties. R must be large while matter exists and finite as $\rho \to 0$ (STUR hypothesis):

```
\Gamma(\rho) = \mathcal{A}\, \exp\!\left[-\frac{B_0 + R(\rho)}{\hbar}\right]
```

The unifying claim is that both are set by Φ_R. The Bekenstein bound links them: as energy density goes to zero, the information a region can hold goes to zero, so the ceiling on Φ_R^2 falls with it. The empty universe is then a zero-information blank slate, which is the stated trigger for tunneling. Writing $R(\rho)$ and ρ_c explicitly in terms of Φ_R is open.

Open problems and falsification targets

The equation stands or falls on these, in order of importance:

- [ ] **Test the postulate $S_R = \frac{1}{2} k_B \Phi_R^2$.** It predicts $\Phi_R^2 = 2E/(k_B T)$ for energy E exchanged: at fixed energy deposited, a colder interferometer loses more visibility. First check whether standard decoherence theory already predicts this scaling. If yes, the postulate is consistent but not new. If no, it is a falsifiable signature for atom interferometers (MAGIS-100, AION).

- [ ] **Fix the connection A_μ.** Spin connection, U(1), or new field. Until it is fixed, Φ_R has no definite value in any experiment.
- [ ] **Derive $\sigma(\mu)$.** A flow equation carrying the QCD tension to the Planck tension across 38 orders of magnitude, with electromagnetism kept on the perimeter law in between.
- [ ] **Show the two areas are the same kind.** The QCD area law uses loops in spacetime; the black hole area is a horizon. AdS/CFT connects them, but our universe has positive Λ, where holography is not established.
- [ ] **Quark quantum numbers.** Field solitons (Step 6) give baryons. Getting quarks from spacetime folds requires 3 colours, charges ⅔ and −⅓, three generations and confinement. No existing work does this.
- [ ] **Write $R(\rho)$ and ρ_c in terms of Φ_R** so one quantity sets both the bounce and the tunneling barrier.
- [ ] **Information paradox.** If evaporation ends at zero information, STUR must say where black hole information goes.

References

Standard literature, cited from knowledge and not link-checked; verify page details before submission.

- Ashtekar, Pawlowski, Singh, "Quantum nature of the big bang", Phys. Rev. Lett. 96, 141301 (2006)
- Battye, Sutcliffe, "Symmetric Skyrmions", Phys. Rev. Lett. 79, 363 (1997)
- Bekenstein, "Universal upper bound on the entropy-to-energy ratio for bounded systems", Phys. Rev. D 23, 287 (1981)
- de Boer, Verlinde, Verlinde, "On the holographic renormalization group", JHEP 08 (2000) 003
- Derrick, "Comments on nonlinear wave equations as models for elementary particles", J. Math. Phys. 5, 1252 (1964)
- Englert, "Fringe visibility and which-way information: an inequality", Phys. Rev. Lett. 77, 2154 (1996)
- Friedman, Sorkin, "Spin ½ from gravity", Phys. Rev. Lett. 44, 1100 (1980)
- Hawking, "Particle creation by black holes", Commun. Math. Phys. 43, 199 (1975)
- Houghton, Manton, Sutcliffe, "Rational maps, monopoles and Skyrmions", Nucl. Phys. B 510, 507 (1998)

- Landauer, "Irreversibility and heat generation in the computing process", IBM J. Res. Dev. 5, 183 (1961)
- Maldacena, "Wilson loops in large N field theories", Phys. Rev. Lett. 80, 4859 (1998)
- Penrose, Cycles of Time (Bodley Head, 2010)
- Rovelli, Vidotto, "Planck stars", Int. J. Mod. Phys. D 23, 1442026 (2014)
- Ryu, Takayanagi, "Holographic derivation of entanglement entropy from AdS/CFT", Phys. Rev. Lett. 96, 181602 (2006)
- Skyrme, "A unified field theory of mesons and baryons", Nucl. Phys. 31, 556 (1962)
- Smarr, "Mass formula for Kerr black holes", Phys. Rev. Lett. 30, 71 (1973)
- Verlinde, "On the origin of gravity and the laws of Newton", JHEP 04 (2011) 029
- Vilenkin, "Creation of universes from nothing", Phys. Lett. B 117, 25 (1982)
- Witten, "Current algebra, baryons, and quark confinement", Nucl. Phys. B 223, 433 (1983)